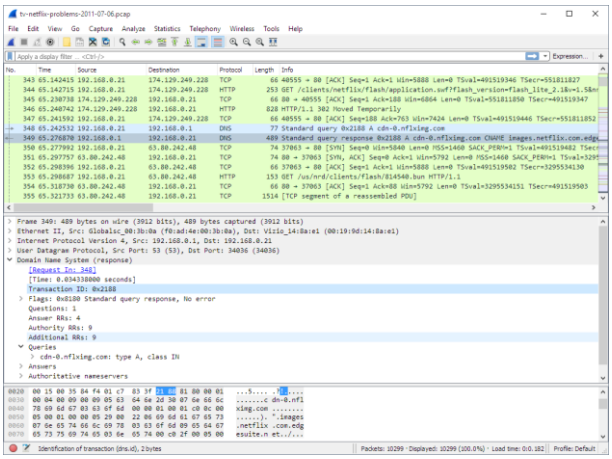
Capture filters vs display filters in Wireshark, with 10 working examples.

1. Introduction to Wireshark



Wireshark is an open-source network protocol analyzer used to capture, inspect, and analyze network packets.

It provides detailed information about packet headers, protocols, addresses, ports, flags, and available payload contents.

Wireshark is widely used for network troubleshooting, security monitoring, incident response, and digital forensics.

2. Capture Filters

A capture filter determines which packets are recorded during a packet capture.

Capture filters use Berkeley Packet Filter (BPF) syntax.

Example:

tcp port 80

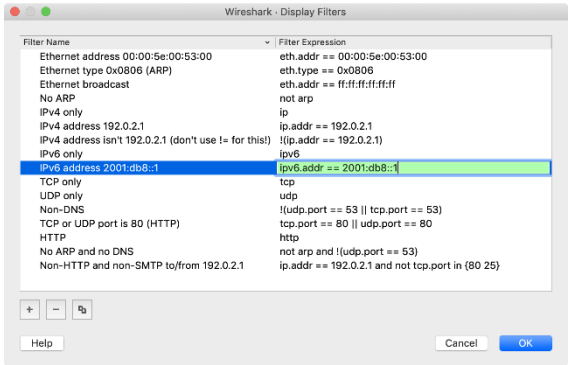
This captures TCP packets involving port 80.

Characteristics

- Applied while capturing traffic.
- Reduces the number of packets stored.
- Helps control capture size and processing overhead.
- Packets excluded by the filter cannot be recovered from that capture.

3. Display Filters

A display filter determines which already captured packets are shown in Wireshark.



Example:

tcp.port == 80

This displays TCP packets involving port 80.

Characteristics

- Applied during packet analysis.

- Does not remove packets from the capture file.
- Supports detailed protocol-field filtering.
- Useful for isolating suspicious communication.

4. Capture Filters vs Display Filters

Feature	Capture Filter	Display Filter
Purpose	Controls packet collection	Controls packet visibility
Applied	During capture	During analysis
Syntax	BPF-style syntax	Wireshark display-filter syntax
Data effect	Excludes nonmatching packets from capture	Hides nonmatching captured packets
Main advantage	Reduces capture size	Enables detailed investigation
Example	tcp port 443	tcp.port == 443

5. Ten Working Filter Examples

No.	Purpose	Capture Filter	Display Filter
1	TCP packets	tcp	tcp
2	UDP packets	udp	udp
3	ICMPv4 packets	icmp	icmp
4	ARP packets	arp	arp
5	DNS traffic	port 53	dns
6	HTTP on TCP port 80	tcp port 80	tcp.port == 80
7	HTTPS on TCP port 443	tcp port 443	tcp.port == 443
8	Specific IPv4 host	host 192.168.1.10	ip.addr == 192.168.1.10
9	SMB on TCP port 445	tcp port 445	tcp.port == 445
10	IPv4 subnet	net 192.168.1.0/24	ip.addr == 192.168.1.0/24

6. Additional Security-Focused Display Filters

Initial TCP SYN packets

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

TCP reset packets

```
tcp.flags.reset == 1
```

DNS NXDOMAIN responses

```
dns.flags.rcode == 3
```

HTTP POST requests

```
http.request.method == "POST"
```

TLS traffic

```
tls
```

SMB2/SMB3 traffic

smb2

Packets from a specific source IP

ip.src == 192.168.1.10

TCP retransmissions identified by Wireshark

tcp.analysis.retransmission

7. Wireshark in Security Analysis

Wireshark helps identify:

- Port scanning and reconnaissance.
- Abnormal TCP connection patterns.
- DNS tunneling indicators.
- ARP spoofing.
- Suspicious HTTP requests.
- Unusual ICMP activity.
- SMB-based lateral movement.
- Unexpected network connections.